



ITI · AI AGENTS DEVELOPMENT USING PYTHON TRACK

Generative AI & Prompt Engineering

Eng. Fatma Elraey*Lecture 1 — Introduction to Generative AI and Prompt Engineering*

Field	Detail
Instructor	Fatma Elraey
Course	Introduction to Generative AI and Prompt Engineering (1-Day Course)
Session	Lecture — 3 Hours
Focus	A beginner-friendly, no-coding-required look at what Generative AI is, the tools built on it, and how to talk to it well through prompt engineering.

Welcome!

You do not need any programming background for this course. By the end of today you will understand what Generative AI actually is, know the key terms people use when talking about it, have tried several real AI tools, and be able to write clear, effective prompts using a simple framework — a skill that transfers to any AI tool you use afterward, in any track you study next.

Today's Outline

1. From AI to Generative AI
2. Key Terms & How LLMs Work
3. Reasoning Models & AI Agents
4. Popular Gen AI Tools & New Features
5. Prompt Engineering Techniques
6. Limitations, Security & Privacy

01

From AI to Generative AI

Artificial Intelligence (AI) is the broad idea of machines performing tasks that normally need human intelligence — recognizing a face, translating a sentence, recommending a movie. Generative AI is a more recent, more specific branch: instead of just recognizing or predicting, it creates brand-new content — text, images, code, audio, even video — that did not exist before.

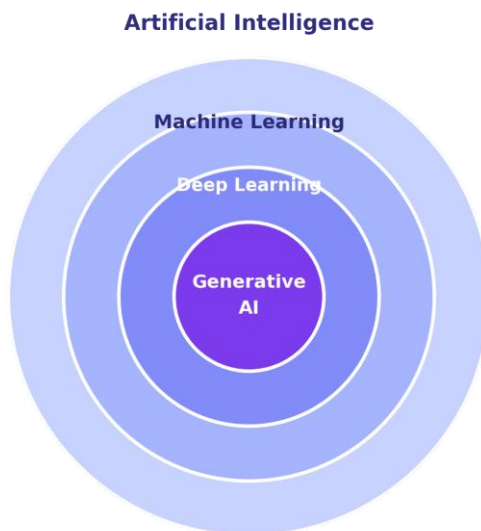


Figure 1.1 — Generative AI is a modern branch inside the wider field of AI.

Why has this suddenly become such a big deal? A few practical reasons: it saves real time on everyday writing and research tasks, it lowers the bar for producing a first draft of almost anything, and it's now available for free or cheaply through a normal web browser — no setup, no installation. Because of that, being comfortable working with AI tools is quickly becoming as basic a skill as being comfortable with a search engine or a spreadsheet.

A quick, honest look at **common uses** today: drafting emails and reports, summarizing long documents, brainstorming ideas, writing and explaining code, generating images and presentation content, and acting as a tutor that can answer follow-up questions.

PRO TIP Whatever track you take after this course, the way you talk to an AI tool barely changes. Learn it once here, and you'll reuse it everywhere — in coding, in data work, in writing, in research.



✓ QUICK CHECK — From AI to Generative AI

Q1. What makes Generative AI different from AI in general?

Answer: *It creates new content (text, images, code, audio) rather than just recognizing patterns or making a prediction.*

Q2. Name two everyday reasons Gen AI has become so widely used.

Answer: *It saves time on everyday writing/research tasks, and it's easy and cheap (often free) to access through a browser, with no setup.*

02 Key Terms & How LLMs Work

Most of the Generative AI tools you'll use for text are powered by a Large Language Model, or **LLM** — a model trained on huge amounts of text that has learned the patterns of language well enough to predict, one small piece at a time, what text should come next.

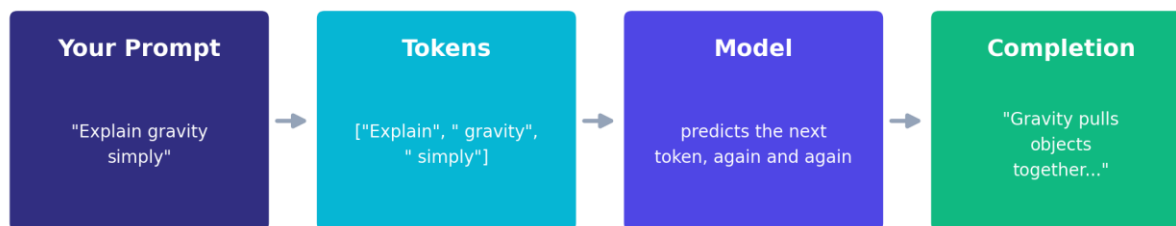


Figure 1.2 — Your prompt becomes tokens, the model predicts the next token again and again, and the result is the completion.

A handful of terms come up constantly, so let's define them plainly, one at a time:

Prompt → the input you give the model (your question or instruction)
Completion → the text the model generates in response
Token → a small chunk of text the model reads and writes
Model → the trained system itself (e.g. GPT-4, Gemini, Claude)

A few more advanced terms are worth simply recognizing, even without needing to master them today:

Multi-modal → a model that can handle more than one type of input, e.g. text AND images, not just text
RAG → "Retrieval-Augmented Generation" -- the model looks up real documents or data before answering, instead of relying only on what it memorized during training
Fine-tuning → further training a general model on a specific, narrower dataset so it gets better at one particular kind of task

PRO TIP You don't need to memorize every term today. Recognizing them when you hear them — and knowing roughly what each one means — is enough at this stage.



✓ QUICK CHECK — Key Terms & How LLMs Work

Q1. What is a token, roughly speaking?

Answer: *A small chunk of text (often part of a word) that the model reads and generates, one at a time.*

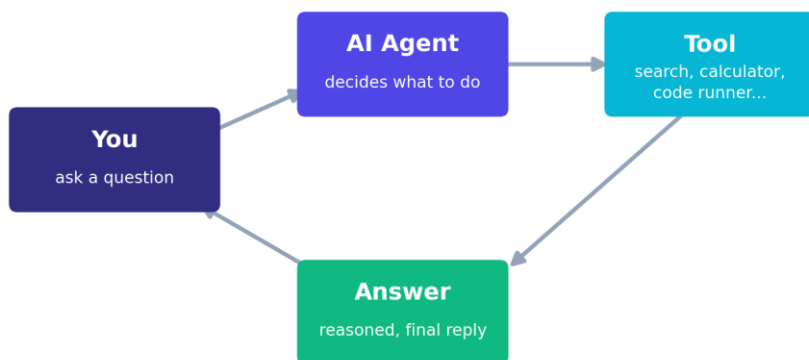
Q2. What does RAG let a model do that it couldn't do from training alone?

Answer: *Look up real documents or data before answering, instead of relying only on what it memorized during training.*

03 Reasoning Models & AI Agents

Some newer models are built (or specifically prompted) to "think before they answer" — working through a problem in steps internally before giving a final response, rather than answering instantly. These are often called reasoning models, and they tend to do noticeably better on harder problems like multi-step math, logic puzzles, or planning tasks, at the cost of taking a little longer to respond.

An AI **agent** takes this a step further: instead of just answering from what it already knows, an agent can decide to use outside tools — searching the web, running a calculation, executing code — and then combine what it finds into a final answer.



The agent can loop through tools more than once before answering.

Figure 1.3 — An agent can call a tool, look at the result, and decide what to do next.

This matters because it changes what these tools can reliably do. A plain **model** can only work from what it learned during training and whatever you tell it in the prompt. An **agent** can go check something first — today's weather, a live stock price, the contents of a specific file — before answering, which makes its answers far more current and grounded.

PRO TIP At this introductory level, you don't need to build an agent yourself — just recognize the idea when a tool mentions "agent mode" or "tools," and understand it means the AI can take real actions, not just talk.

✓ QUICK CHECK — Reasoning Models & AI Agents

Q1. What's the general idea behind a 'reasoning model'?

Answer: It works through a problem in steps internally before giving a final answer, rather than responding instantly — which tends to help on harder, multi-step problems.

Q2. What can an AI agent do that a plain model cannot?

Answer: It can use outside tools (search, calculators, code execution, etc.) to gather fresh information before answering, instead of relying only on what it already knows.

04 Popular Gen AI Tools & New Features

There is no single "correct" Generative AI tool — several major companies each offer their own, and most do similar things reasonably well. It's worth knowing the landscape rather than committing to just one.

A few well-known Generative AI tools



Figure 1.4 — A few well-known Generative AI tools you'll likely encounter.

Rather than memorizing every feature of every tool, it's more useful to know the categories of capability that keep showing up across tools, since they tend to appear under different names in different products:

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Deep Search / Research mode → the AI browses multiple sources
                             and compiles a researched, cited answer
Agent mode                  → the AI can take multi-step actions on its
                             own -- browsing, running code, and more
Learn / Study mode          → instead of just giving the answer, the AI
                             guides you toward it, like a tutor
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As one concrete example: ChatGPT's "Study" mode intentionally holds back the full answer and instead asks guiding questions, which is genuinely useful while you're still learning a topic and want to think it through yourself rather than just copy an answer.

PRO TIP Try the same question in two different tools before trusting either answer fully — the tools genuinely disagree sometimes, and comparing them is one of the fastest ways to build good judgment about when to trust an AI's response.

✓ QUICK CHECK — Popular Gen AI Tools & New Features

Q1. What does a 'Deep Search' or 'Research' mode typically do differently from a normal chat answer?

Answer: *It browses multiple sources and compiles a researched, cited answer, instead of answering purely from what the model already knows.*

Q2. What is the point of a 'Study' or 'Learn' mode?

Answer: *Instead of just giving the final answer, it guides you toward it with questions, acting more like a tutor than an answer machine.*

05 Prompt Engineering Techniques

Prompt engineering is simply the skill of writing clear instructions so an AI tool gives you a better, more useful answer. It's less about a special trick and more about being as clear with the AI as you would be giving instructions to a new team member who's eager to help but knows nothing about your situation yet.

A **vague** prompt gets a **vague** answer:

PROMPT

Write about marketing.

RESPONSE (shortened)

Marketing is the process of promoting and selling products or services...
(generic, unfocused, could go anywhere)

A **specific** prompt gets a **specific, useful** answer:

PROMPT

Write 3 Instagram captions for a small coffee shop's new pumpkin spice latte, friendly and under 15 words each.

RESPONSE (shortened)

1. Fall just got a little cozier.
2. Pumpkin spice season is officially here!
3. Your new favorite latte is waiting.

A simple, reliable framework for building a good prompt is R-T-C-F — Role, Task, Context, Format. You don't need all four every time, but reaching for them when a prompt isn't working well almost always helps:

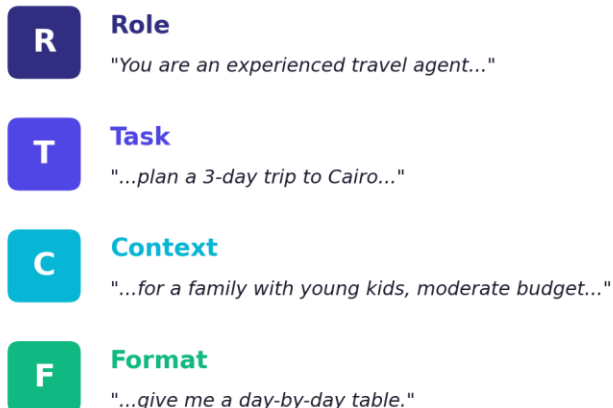


Figure 1.5 — Role, Task, Context, and Format — a simple framework for a stronger prompt.

Two more techniques worth knowing by name, since you'll see them referenced often:



Few-shot prompting → giving the AI 1-2 examples of the format you want, before asking for the real thing

Chain-of-thought → asking the AI to "think step by step" or "show your reasoning" before giving a final answer

PRO TIP *If a response disappoints you, don't start over — refine it. Tell the AI exactly what was wrong ("too long," "too formal," "missing X") and ask it to try again. Iterating is normal and usually faster than writing the perfect prompt on the first try.*

✓ QUICK CHECK — Prompt Engineering Techniques

Q1. What do the four letters in the R-T-C-F prompt framework stand for?

Answer: Role, Task, Context, Format.

Q2. What is few-shot prompting?

Answer: Giving the AI one or two examples of the format or style you want before asking for the real thing.

Q3. What is the fastest way to improve a disappointing AI response?

Answer: Tell it specifically what was wrong and ask it to try again, rather than starting over from scratch.

06 Limitations, Security & Privacy

As useful as these tools are, they are not perfect, and knowing their weak points is just as important as knowing their strengths.

Known Limitations

Hallucination → the AI can state something false confidently and fluently, with no warning that it might be wrong
Outdated info → a model's training data has a cutoff date -- it may not know about very recent events unless it can search
Bias → the model can reflect biases present in its training data
Math & precision → plain language models can make careless arithmetic mistakes, especially with large numbers

The practical takeaway: always double-check anything factual, numerical, or high-stakes before relying on it — treat an AI's answer as a strong first draft, not a guaranteed final answer.

Security & Data Privacy

- Don't paste passwords, API keys, or other secrets into a prompt.
- Assume anything you type may be stored or reviewed -- avoid sharing confidential company data or other people's personal information.
- Check your organization's policy before using AI tools on work data.
- Be cautious with AI-generated code or configuration before running it -- review it the same way you'd review a colleague's code.

PRO TIP *A good habit: before pasting anything into an AI tool, ask yourself "would I be comfortable if this text were posted publicly?" If not, don't paste it.*

That wraps up the lecture portion. In the lab, you'll get hands-on and put everything from today — clear prompts, the R-T-C-F framework, few-shot examples, and comparing tools — into practice.

✓ QUICK CHECK — Limitations, Security & Privacy

Q1. What is 'hallucination' in the context of an AI model?

Answer: *When the AI states something false confidently and fluently, with no obvious warning sign that it might be wrong.*

Q2. What's a simple rule of thumb for what's safe to paste into an AI prompt?

Answer: *If you wouldn't be comfortable with that text being posted publicly, don't paste it in — this covers passwords, secrets, and confidential or personal data.*

☆ Check Your Understanding

Work through these on your own before checking the marked answer.

Practice Question 1

What best describes the relationship between AI, Machine Learning, and Generative AI?

- A. They are three unrelated fields
- B. Generative AI is the broadest field, containing all of AI
- C. Generative AI is a specific, modern branch nested inside the broader fields of AI and Machine Learning
- D. Machine Learning came after Generative AI

Practice Question 2

Which of these is the best definition of a 'token' in the context of an LLM?

- A. An entire document given to the model
- B. A small chunk of text (often part of a word) the model reads or generates
- C. A password used to access the model
- D. The final answer the model gives

Practice Question 3

A tool searches the web, checks a live calculator, and combines the results into an answer. What is this tool acting as?

- A. A plain chat model with no extra capability
- B. An AI agent using tools
- C. A fine-tuned model only
- D. This isn't possible for any AI tool

Practice Question 4

Which prompt is more likely to get a specific, useful response?

- A. "Tell me about dogs."
- B. "Write a 100-word intro paragraph for a blog post aimed at first-time dog owners, in a warm and encouraging tone."
- C. "Dogs?"
- D. Both are equally likely to work well

Practice Question 5

What does 'hallucination' mean when talking about an AI model's answers?

- A. The model refuses to answer
- B. The model states something false, confidently and fluently, without flagging uncertainty
- C. The model only answers in images
- D. The model is too slow to respond

Practice Question 6

Before pasting a block of text into an AI chat tool, what's the recommended quick privacy check?

- A. Check if the text is longer than 500 words
- B. Ask yourself whether you'd be comfortable if that text were posted publicly
- C. Translate it into another language first
- D. There's no need to check anything

Answers

Practice Question 1

What best describes the relationship between AI, Machine Learning, and Generative AI?

- A. They are three unrelated fields
- B. Generative AI is the broadest field, containing all of AI
- C. Generative AI is a specific, modern branch nested inside the broader fields of AI and Machine Learning ✓**
- D. Machine Learning came after Generative AI

Why: AI is the broadest field; Machine Learning is a major approach within it; Generative AI is a more recent, more specific branch that focuses on creating new content.

Practice Question 2

Which of these is the best definition of a 'token' in the context of an LLM?

- A. An entire document given to the model
- B. A small chunk of text (often part of a word) the model reads or generates ✓**
- C. A password used to access the model
- D. The final answer the model gives

Why: Models don't read or write whole words or sentences at once — they process small chunks of text called tokens, one at a time.

Practice Question 3

A tool searches the web, checks a live calculator, and combines the results into an answer. What is this tool acting as?

- A. A plain chat model with no extra capability
- B. An AI agent using tools ✓**
- C. A fine-tuned model only
- D. This isn't possible for any AI tool

Why: Using outside tools (search, calculators, code) and combining the results into a final answer is the defining behavior of an AI agent.

Practice Question 4

Which prompt is more likely to get a specific, useful response?

- A. "Tell me about dogs."
- B. "Write a 100-word intro paragraph for a blog post aimed at first-time dog owners, in a warm and encouraging tone." ✓**
- C. "Dogs?"
- D. Both are equally likely to work well

Why: The second prompt gives a role/context (first-time dog owners), a clear task (a 100-word intro paragraph), and a format/tone (warm and encouraging) — the R-T-C-F pieces that make a prompt specific.

Practice Question 5

What does 'hallucination' mean when talking about an AI model's answers?

- A. The model refuses to answer
- B. The model states something false, confidently and fluently, without flagging uncertainty ✓**
- C. The model only answers in images
- D. The model is too slow to respond

Why: Hallucination specifically refers to the model producing incorrect information that sounds just as confident and fluent as correct information — which is exactly what makes it risky to trust blindly.

Practice Question 6

Before pasting a block of text into an AI chat tool, what's the recommended quick privacy check?

- A. Check if the text is longer than 500 words
- B. Ask yourself whether you'd be comfortable if that text were posted publicly ✓**
- C. Translate it into another language first
- D. There's no need to check anything

Why: This simple public-comfort test naturally rules out passwords, secrets, confidential company data, and other people's personal information.